

Ready-Mix Concrete (RMC) Data Standard

Mix Proportions (M10 & M20), Quality Control Protocols & Production Guidelines (IS 4926 / IS 10262)

1. Standard Concrete Mix Proportions (Per 1 m³)

Standard mix design parameters for unreinforced levelling concrete (M10) and reinforced structural concrete (M20).

Mix Property / Material	M10 Grade (PCC / Levelling)	M20 Grade (RCC / Structural)
Nominal Mix Ratio	1 : 3 : 6	1 : 1.5 : 3
Characteristic Strength (f_{ck})	10 N/mm² @ 28 days	20 N/mm² @ 28 days
Target Mean Strength (f'_{ck})	13.5 N/mm²	26.5 N/mm²
Cement Content (OPC 43/53)	220 kg/m³	320 kg/m³
Fine Aggregate (Sand / M-Sand)	750 kg/m³	680 kg/m³
Coarse Aggregate (20 mm)	700 kg/m³	660 kg/m³
Coarse Aggregate (10 mm)	480 kg/m³	440 kg/m³
Water Content	130 L/m³	160 L/m³
Water-Cement Ratio (w/c)	Max 0.60	Max 0.50
Chemical Admixture (Plasticizer)	Optional (0.2% – 0.5%)	0.5% – 1.0% by cement weight

2. Quality Control & Testing Specifications

A. Workability & Slump Parameters (IS 1199)

- **Plant Target Slump:** 120 mm ± 25 mm
- **Site Placement Slump:** 75 – 100 mm (Pumpable)
- **Slump Retention:** 2 to 3 hours (using retarding plasticizers)
- **Temperature Control:** Max concrete temp at delivery ≤ 32°C

B. Compressive Strength Acceptance (IS 516)

- **7-Day Strength Target:** ≥ 65% of specified f_{ck}
- **28-Day Strength Target:** ≥ 100% of specified f_{ck}
- **Sampling Rate:**
 - 1 set (3 cubes) for 1–5 m³
 - 2 sets for 6–15 m³
 - 4 sets for 16–30 m³ (+1 set per additional 50 m³)

3. Production Standards & Batching Tolerances (IS 4926)

Batching Plant Accuracy & Transit Mixer Operation

- **Batching Tolerances:** Cement & Fly Ash: $\pm 1\%$ | Water & Admixtures: $\pm 1\%$ | Aggregates: $\pm 2\%$
- **Mixing Time:** Minimum 30 to 45 seconds per batch in twin-shaft plant mixers.
- **Agitating Speed (En Route):** 2 – 6 RPM during transport.
- **Mixing Speed (Site/Plant Re-mixing):** 12 – 18 RPM for 3 to 5 minutes prior to discharge.
- **Maximum Discharge Time:** Complete discharge within **2 hours** of batching (or 1.5 hours if ambient temp > 30°C).

4. Database / API Import Schema (JSON Format)

Below is the structured JSON object layout recommended for importing these standards into your application database:

```
{
  "rmc_standards": {
    "mix_designs": [
      {
        "grade": "M10",
        "type": "PCC",
        "target_fck_28days_N_mm2": 10.0,
        "target_mean_strength_N_mm2": 13.5,
        "max_w_c_ratio": 0.60,
        "mix_proportions_kg_m3": {
          "cement": 220,
          "fine_aggregate": 750,
          "coarse_agg_20mm": 700,
          "coarse_agg_10mm": 480,
          "water_liters": 130,
          "admixture": 0.0
        }
      },
      {
        "grade": "M20",
        "type": "RCC",
        "target_fck_28days_N_mm2": 20.0,
        "target_mean_strength_N_mm2": 26.5,
        "max_w_c_ratio": 0.50,
        "mix_proportions_kg_m3": {
          "cement": 320,
          "fine_aggregate": 680,
          "coarse_agg_20mm": 660,
          "coarse_agg_10mm": 440,
          "water_liters": 160,
          "admixture": 2.56
        }
      }
    ],
    "tolerances": {
      "cement_percent": 1.0,
      "water_percent": 1.0,
      "aggregates_percent": 2.0
    }
  }
}
```

```
}  
}  
}
```